Speech Science - Lecture 2: Speech Science (19 Oct)

Lecture overview

This lecture introduces Speech Science 2, focusing on the acoustic analysis of speech signals through time-domain and frequency-domain representations. The time wave (oscillogram) visualizes sound pressure over time, revealing periodic patterns in vowels and nasals, aperiodic noise in fricatives, and transient bursts in plosives. Fundamental frequency (F0), representing the rate of vocal fold vibration, is calculated by averaging multiple periods to account for natural cycle variations. The lecture then transitions to spectral analysis via Source-Filter theory, explaining how the glottal source spectrum is shaped by the vocal tract acting as a bandpass filter. This interaction creates distinct formant frequencies (resonances), with the first three formants (F1-F3) being critical for identifying vowel identity based on the vocal tract's length and shape rather than the sound source itself.

The relationship between acoustics and physiology is illustrated through vowel space charts, such as the Peterson and Barney study, which map F1 against F2 frequencies to distinguish vowel qualities across different speakers. F1 primarily correlates with vowel height (openness), while F2 distinguishes front from back tongue positions, forming a quadrilateral vowel space that varies between languages like Dutch and American English. Specific acoustic signatures define vowel categories: front vowels exhibit large F1-F2 separation, back vowels show small F1-F2 separation, and central vowels display equally spaced formants. Additionally, the lecture addresses temporal properties, noting that vowel duration is not fixed but influenced by stress, speaking rate, and adjacent consonants, with diphthongs characterized by dynamic formant transitions rather than steady states.

Slide 1: Speech Science 2 2026-2027

This course is titled Speech Science 2 for the 2026-2027 academic year. It is led by Professor Astrid van Wieringen from the Department of Neurosciences at KU Leuven, where she holds a position in Experimental Otorhinolaryngology (ORL).

Slide 2: Experimenta

“Experimental ORL2” is a section or experiment marker. The slide supplies no procedure or result by itself; the following pages carry the acoustic material.

Slide 3: Time (s)

A time wave, or oscillogram, plots sound pressure against time, exemplified by the phrase "The cat caught a mouse." Unlike a spectrum, this representation reveals distinct waveform characteristics: periodic patterns in nasals, vowels, and approximants; aperiodic noise in fricatives; quiescent silence preceding plosives (voice onset time); and transient release bursts. This visualization captures how speech sounds vary in duration and intensity over time.

Slide 4: Time (s)

To calculate fundamental frequency (F0) from a time signal, measure the total duration of multiple periods and divide by the period count to find the average period length (T). For example, five periods of the vowel /a/ spanning 0.03027 seconds yield an average T of 0.00605 s, resulting in an F0 of 165 Hz using the formula f = 1/T. Because F0 varies slightly between individual cycles, averaging several periods provides a more accurate measurement than relying on a single period.

Slide 5: Experimenta

Slide 5 presents experimental data comparing two conditions, labeled A and B, involving the phonetic sequence /s/ /o/ /l/ /d/ /ier/. The slide displays specific time measurements in seconds for each condition: Condition A records a duration of 0.383628 s, while Condition B records 0.140272 s. Both conditions share identical boundary values of -0.8545 and 0.8741, suggesting a comparison of temporal dynamics within the same experimental framework.

Slide 6: Each (vowel) sound has a characteristic spectrum shape depending on formant frequency

Each vowel sound possesses a unique spectrum shape determined by formant frequencies, which act as bandpass filters amplifying specific harmonics from the source. According to Source-Filter theory, the glottal source spectrum naturally decreases at -6 dB per octave due to radiation characteristics, while the vocal tract filter enhances energy bundles near resonant frequencies. This interaction creates distinct spectral coloring, allowing different vowels like 'ah' or 'ooh' to be produced from the same vocal fold vibration by altering the filter shape.

Slide 7: • Usually only the first three formants are considered. Exact values depend on length and shape of VT (place of constriction and

While five formants exist, analysis typically focuses on the first three (F1-F3), with F1 and F2 being critical for identifying vowel identity and F3 contributing to quality. These resonant frequencies appear as spectral peaks determined by the vocal tract's length, shape, and constriction rather than the sound source itself. Unlike the single-value fundamental frequency, formants represent bands of energy where specific harmonics are amplified or attenuated by the vocal tract filter to create distinct speech sounds.

Slide 8: Experimenta

Slide 8 introduces "Experimenta," specifically focusing on the perceptual, physiologic, and acoustic aspects of ORL 8. The professor's instruction to "Listen" emphasizes that these concepts are fundamentally grounded in auditory perception and require active listening to understand the relationship between physical sound properties and human physiological response.

Slide 9: Relation between acoustics and physiology

Slide 9 illustrates the relationship between acoustics and physiology by plotting F1 versus F2 frequencies for vowels, highlighting significant variation among speakers as shown in the Peterson and Barney (1952) study. The professor clarifies that while individuals produce the same vowel sound, their fundamental frequency differs physiologically based on vocal fold vibration rates; for example, a male's 100 Hz voice corresponds to 100 vibrations per second. This physiological source mechanism, driven by air pressure through the vocal folds, underpins the acoustic variations observed across different speakers.

Slide 10: 16

This scatter plot displays the first and second formant frequencies of Dutch vowels from Belgian speakers, illustrating how acoustic properties map to articulatory features. The vertical F1 axis distinguishes vowel height, ranging from open/low /a/ to close/high /i/, while also indicating rounding differences between /u/ and /a/. The horizontal F2 axis differentiates front vowels like /i/ from back vowels like /u/, demonstrating the systematic relationship between formant values and vowel quality.

Slide 11: 17

Slide 17 presents the first and second formants (F1 and F2) for eight cardinal American-English vowels, citing Borden and Harris (1984). The professor clarifies that these vowel positions are mapped onto a quadrilateral chart rather than a triangle, organized by the openness or closeness of the sounds. This geometric representation serves as a guide within the International Phonetic Alphabet to assist transcribers in accurately locating and distinguishing various vowel qualities.

Slide 12: 18

This 1952 Peterson and Barney study illustrates the relationship between acoustics and physiology by plotting the first (F1) versus second (F2) resonant frequencies for ten American English vowels across 76 speakers. The resulting vowel space forms a quadrilateral shape, distinct from the Dutch system, and notably includes two separate /a/ variants: /ɑ/ as in "but" and /æ/ as in "want." The data highlights significant variation in frequency estimation among individuals while mapping specific phonetic symbols like /u/, /o/, /ɛ/, /e/, /ɪ/, and /i/.

Slide 13: Front vowels /i, I, e, ɛ, □/

Front vowels /i, I, e, ɛ, □/ are characterized by a front tongue position and acoustic cues showing large F1-F2 separation with close F2-F3 positioning. While higher tongue positions yield lower F1 values, long tense vowels (/i, e, □/) possess longer inherent duration than short vowels (/I, ɛ/), though the professor notes this distinction is not always discrete in actual measurements. Additionally, some high vowels exhibit higher intrinsic fundamental frequency (F0).

Slide 14: Phonemes /u, ʊ, o, ɔ, ɑ/

The back vowels /u, ʊ, o, ɔ, ɑ/ are characterized by a back tongue position and varying heights, acoustically distinguished by a small F1-F2 separation and large F2-F3 separation. These phonemes vary primarily in F1 frequency, where a lower tongue position corresponds to a higher F1. Temporally, /u, o, ɔ, ɑ/ possess inherently long durations, whereas /ʊ/ is short.

Slide 15: Phonemes /ʌ, ə/

The phonemes /ʌ/ and /ə/ are central vowels characterized by a central tongue position and an acoustic uniform formant pattern where frequencies are equally spaced. A key temporal distinction is that /ʌ/ has a longer duration than /ə/. These spectral and temporal properties summarize the behavior of central vowels as documented in linguistic research.

Slide 16: • No fixed value for vowel duration (inherent duration). Duration is influenced by:

Vowel duration lacks a fixed inherent value and is instead influenced by factors such as syllable stress, speaking rate, the voicing and place of articulation of adjacent consonants, utterance position, and word familiarity. While individual speaking rates vary, vowels are uniquely capable of extension compared to transient sounds like stops, with diphthongs inherently lasting longer than monophthongs. Understanding these variable parameters is fundamental for accurate acoustic measurement and exam preparation.

Slide 17: • Diphthongs /ai/, /au/:

Diphthongs /ai/ and /au/ are produced with a relatively open vocal tract, characterized by a dynamic formant structure that transitions between steady states rather than maintaining a single pattern. Unlike stationary monophthongs, these sounds involve a glide from an on-glide to an off-glide, exhibiting slow dynamic changes where the specific rate of frequency change is often more invariant and perceptually significant than the exact onset or offset values.